

FRONTEND • BACKEND • DATABASE

FRONTEND — Web / PWA

HTML · CSS · JavaScript | folder web/

- Farmer screens: Photo, Field, News, Shop, Call, Help, chat
- Camera / gallery = HTML file input (no Capacitor Camera plugin)
- Bangla / English, login, vendor desk, admin desk
- PWA service worker sw.js

BACKEND / API — FastAPI

backend/main.py · Uvicorn. Frontend calls:

- POST /api/predict — photo → leaf → crop → disease
- POST /api/chat, /chat/vision, /chat/stt
- Farmer auth /api/auth/* · vendor /api/vendor/*
- Admin /api/admin/* · shop, news, advice, farm APIs

DATA / STORAGE

Server-side application data



- users.db — farmer accounts
- agroscan_shop.db — shop & vendors
- JSON pack — 142 disease cards
- *.pt — PyTorch CNN weights

DATABASE NOTE

The backend DBMS is **SQLite**. Farmer sessions: data/users.db. Shop / vendor / admin: data/agroscan_shop.db.

The frontend also uses browser **localStorage** for cart, scan history, field plots, language, and the session token. That is not the server database.

CNN weights are files, not rows: leaf_gate.pt, leaf_type.pt, disease_*.pt.

Live: agroscan.mujahidulhaquejihad.com

MOBILE APP — Capacitor Android

Capacitor 6.2 | com.mujahidulhaquejihad.agroscan

- Same web UI copied into mobile/www
- No on-device model — scan, chat, shop call the hosted API
- Native plugins: Geolocation, App, Status Bar
- APK built with Gradle; API base = production URL

AI + APPLICATION SERVICES

Runs on the server, only through those APIs

- L1 MobileNetV3-Large → L2 EfficientNet-B3 → L3 ensemble (B3 + ResNet-50 + DenseNet-121), crop-masked, highest confidence wins
- Advice comes from the JSON pack, not from the CNN
- Chat: clarify → pack → KB → Gemini + TF-IDF RAG (k = 6)
- Shop orders, vendor stock, admin users
- Weather is not this API — the page calls Open-Meteo in the browser

Farmer / vendor / admin → Web or Capacitor APK → FastAPI → SQLite + JSON pack + .pt models

